

Lab 7: Evaluation & Optimization

Evaluation: 5% of the total grade

In this lab exercise you will work with the digits dataset in sklearn which consists of features from images of hand-written digits from 0-9. The images attribute of the dataset stores 8x8 arrays of grayscale values for each image. There are 64 descriptive features and 10 classes in the target feature dataset. Details of the dataset can be found [here](#).

The objective is to predict the class of the image based on the descriptive features.

Your task is to train and evaluate three (3) classification models with different algorithms (SVM, Logistic regression, & KNN) and compare performance on the test dataset using multiple measures of performance as discussed in the class.

Perform the following tasks:

1. Train and evaluate three (3) classification models with SVM, Logistic regression, & KNN
2. Select the best performer based on measures of performance you learned in the class
3. Perform hyperparameter tuning to optimize the performance of the selected model using any two of the following techniques:
 - a) Grid Search
 - b) Random Search
 - c) Bayesian Optimization
4. Briefly summarize the performance of the classification models and hyperparameter tuning techniques.

Submit your lab report along with the source code preferably as Jupyter notebook in pdf format by the due date as schedule on the Nexus.

Lab Grading Criteria

Fraction	0.8 – 1	0.4 – 0.7	0 – 0.3	Max Score
Category				
Model development workflow	Followed all steps in the modeling process	Followed most steps in the modeling process	No order in the modeling process	1
Tasks and observations	Performed all the tasks correctly, results properly presented and well explained	Performed most of the tasks correctly, results fairly presented and explained	Performed some of the tasks correctly, poorly presented results with little explanation	3
Code	Well organized, easy to follow and clean code with meaningful comments	Somewhat organized, a little hard to follow and fairly clean code with some comments	Not organized, very hard to follow and messy code with little or no comments	1
Total				5